# Ogbonna Arinze

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## SUMMARY

Senior Full Stack Engineer with 8 years of professional experience delivering end-to-end product features across fintech, healthcare, civic tech, and enterprise environments. Deep expertise in React, TypeScript, and Next.js on the frontend, with proven backend capabilities in NodeJS, NestJS, PostgreSQL, and RESTful API design. Track record of building financial analytics dashboards with real-time data visualization, architecting scalable systems with Docker and CI/CD pipelines, and driving engineering quality through automated testing, code reviews, and performance optimization. Product-oriented collaborator with experience in high-volume transaction processing and data-intensive applications.

## CORE COMPETENCIES

- **Full Stack Development:** End-to-end feature delivery spanning React/Next.js frontends, NodeJS/NestJS backends, RESTful API design, and relational/NoSQL databases.
- **Frontend Frameworks:** React, Next.js, Vue.js, Redux, Zustand, React Query, Radix UI, Tailwind CSS, and Storybook.
- **Backend & Data:** NestJS, NodeJS, TypeScript, PostgreSQL (with pgvector), MongoDB, Prisma, GraphQL, JWT authentication, and RBAC.
- **Data Visualization & Real-Time Systems:** Recharts, react-chartjs, RxJS, Server-Sent Events (SSE), EventSource API, and financial analytics dashboards.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Cloud & Infrastructure:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Nx monorepo orchestration, and Traefik reverse proxy.
- **Performance & Observability:** Core Web Vitals, code splitting, CDN strategies, Lighthouse, Sentry, Datadog RUM, Langfuse, and bundle analysis.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.
- **Leadership:** Technical architecture reviews, mentorship, code reviews, Agile/Scrum, TDD/BDD, and product-minded development.

## WORK EXPERIENCE

**Genentech (Roche)** San Francisco, CA (Remote)

***Frontend Engineer*** *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, delivering end-to-end features spanning UI components, authentication flows, and backend integrations.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to query their data in real time via a conversational interface.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling, client-side route protection, and permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews.

**ActivEdge Technologies** Lagos, Nigeria

***Senior React Developer*** *Oct 2019 - Sep 2021*

- Played a pivotal role in a team of 3 frontend engineers to build a high-volume **bank Settlement Solution**, significantly enhancing transaction processing capabilities to handle thousands to millions of transactions daily, boosting settlement rates by **80%**.
- Spearheaded the development of critical **financial payment modules** (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Developed a **real-time account reconciliation** solution, boosting the reconciliation of daily transactions by **60%** through innovative transaction processing and data management strategies.

- Enhanced system performance and user experience by adopting the native browser **EventSource API** for **real-time data streaming**, replacing conventional socket.io methods for faster updates and a more responsive interface.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, providing real-time financial analytics and streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

**Descasio** Lagos, Nigeria (Hybrid)
***Senior Frontend Developer*** *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improved release confidence.
- Established engineering standards for code quality and deployment using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

**Founders Lair** Beijing, China (Remote)
***Full Stack Engineer (Contract)*** *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance and structured task management.

**SureTrade Business Solutions** Nigeria
***Frontend Developer*** *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

## PROJECTS

**OurNigeria** ournigeria.ng
***Senior Engineer*** *Feb 2026 - Present*

*AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.*

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with **22 interactive chart types** built with **Recharts**, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and built **RESTful APIs** in NestJS with role-based access control and phone OTP authentication.
- Built a **multi-agent AI system** using Mastra with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining pgvector + BM25 full-text search with Reciprocal Rank Fusion and Cohere reranking.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts, and built the Next.js frontend with **React 19**, **Tailwind CSS v4**, dark mode, and responsive mobile layouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing.

## EDUCATION

**University of Nigeria** Nsukka, Enugu
*BSc Computer Science - 4.1 CGPA* *Graduation Date: Sep 2019*